

Step-by-Step WordPress Deployment on AWS EC2 instance.

Introduction

This project is a step-by-step guide to setting up and running a WordPress website on a server. It uses Apache as the web server and MariaDB as the database. The guide covers everything from preparing the server, installing the needed software, setting up WordPress, and finally making the website live on the internet. By following it, you can create a secure, reliable, and ready-to-use WordPress site.

Requirements

Before deploying the WordPress website, ensure the following are installed and configured:

Linux Server – Amazon Linux.

Web Server – Apache installed and running.

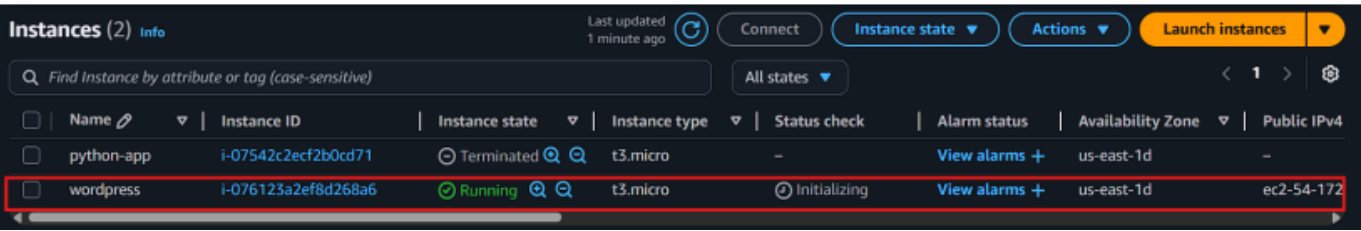
PHP – Along with required PHP extensions.

Database – MariaDB for storing WordPress data.

Steps for Deployment

Step 1: Launch EC2 instance and Establishing a secure connection to your EC2 instance

1. Launch instance



Instances (2) Info								
Last updated 1 minute ago								
Find Instance by attribute or tag (case-sensitive) All states								
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	python-app	i-07542c2ecf2b0cd71	Terminated	t3.micro	-	View alarms +	us-east-1d	-
☐	wordpress	i-076123a2ef8d268a6	Running	t3.micro	Initializing	View alarms +	us-east-1d	ec2-54-172

2. Copy the SSH command

3. Paste command in Git bash

Step 2: Automating LAMP Stack Setup on AWS EC2

- ## 1. Create a LAMP.sh file

```
sudo vim LAMP.sh
```

2. Insert the code for installing apache, mysql and php

```
sudo yum update
```

```
sudo yum install httpd mariadb105-server php -y
```

```
sudo systemctl start httpd mariadb php-fpm
```

```
sudo systemctl enable httpd mariadb php-fpm
```

```
ec2-user@ip-172-31-23-131:~  
sudo yum update  
sudo yum install httpd mariadb105-server php -y  
sudo systemctl start httpd mariadb php-fpm  
sudo systemctl enable httpd mariadb php-fpm  
~  
~
```

3. Run the file

```
ec2-user@ip-172-31-23-131:~  
[ec2-user@ip-172-31-23-131 ~]$ sudo bash LAMP.sh  
Amazon Linux 2023 Kernel Livepatch repository  
Dependencies resolved.  
Nothing to do.  
Complete!  
Last metadata expiration check: 0:00:01 ago on Sun Sep 7 08:28:  
Dependencies resolved.  
=====
```

Package	Architecture
---------	--------------

```
=====
```

Installing:

Step 3: Download and Configure WordPress

#Download WordPress

```
sudo wget https://wordpress.org/latest.tar.gz
```

#Extract the archive

```
sudo tar -xvzf latest.tar.gz
```

```

ec2-user@ip-172-31-23-131:/var/www/html
[ec2-user@ip-172-31-23-131 ~]$ cd /var/www/html/
[ec2-user@ip-172-31-23-131 html]$ sudo wget https://wordpress.org/latest.tar.gz
--2025-09-07 08:31:44-- https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26925441 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz          100%[=====]
2025-09-07 08:31:45 (81.2 MB/s) - 'latest.tar.gz' saved [26925441/26925441]

[ec2-user@ip-172-31-23-131 html]$ ls
latest.tar.gz
[ec2-user@ip-172-31-23-131 html]$
[ec2-user@ip-172-31-23-131 html]$ sudo tar -xvzf latest.tar.gz
wordpress/
wordpress/index.php
wordpress/license.txt
wordpress/readme.html
wordpress/wp-activate.php
wordpress/wp-admin/
wordpress/wp-admin/about.php
wordpress/wp-admin/admin-ajax.php

```

Step 4: Remove latest.tar.gz

```
sudo rm -rf latest.tar.gz
```

```
ls
```

```

[ec2-user@ip-172-31-23-131 html]$ ls
latest.tar.gz  wordpress
[ec2-user@ip-172-31-23-131 html]$ sudo rm -rf latest.tar.gz
[ec2-user@ip-172-31-23-131 html]$ ls
wordpress

```

Step 5: Go to the wordpress folder

```
cd wordpress/
```

```

[ec2-user@ip-172-31-23-131 html]$ cd wordpress/
[ec2-user@ip-172-31-23-131 wordpress]$ ls
index.php      wp-activate.php  wp-comments-post.php  wp-cron.php      wp-load.php      wp-settings.php  xmlrpc.php
license.txt    wp-admin         wp-config-sample.php  wp-includes      wp-login.php     wp-signup.php
readme.html    wp-blog-header.php  wp-content            wp-links-opml.php  wp-mail.php      wp-trackback.php
[ec2-user@ip-172-31-23-131 wordpress]$

```

Step 6: Create WordPress Database

1. Generate the username and password.

```
sudo mysql
```

```
alter user root@localhost identified by 'root';
```

```
[ec2-user@ip-172-31-23-131 wordpress]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> alter user root@localhost identified by 'root';
Query OK, 0 rows affected (0.002 sec)
```

2. Login to Mysql (mariadb105-server)

```
sudo mysql -u root -p
```

```
[ec2-user@ip-172-31-23-131 wordpress]$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input state

MariaDB [(none)]> |
```

3. Create Database

```
#Create Database
```

```
create database wordpressdb;
```

```
#Show Database
```

```
show databases;
```

```
MariaDB [(none)]> create database wordpressdb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| wordpressdb |
+-----+
4 rows in set (0.001 sec)
```

Step 7: Install Connector

```
sudo yum install php8.4-mysqlnd.x86_64
```

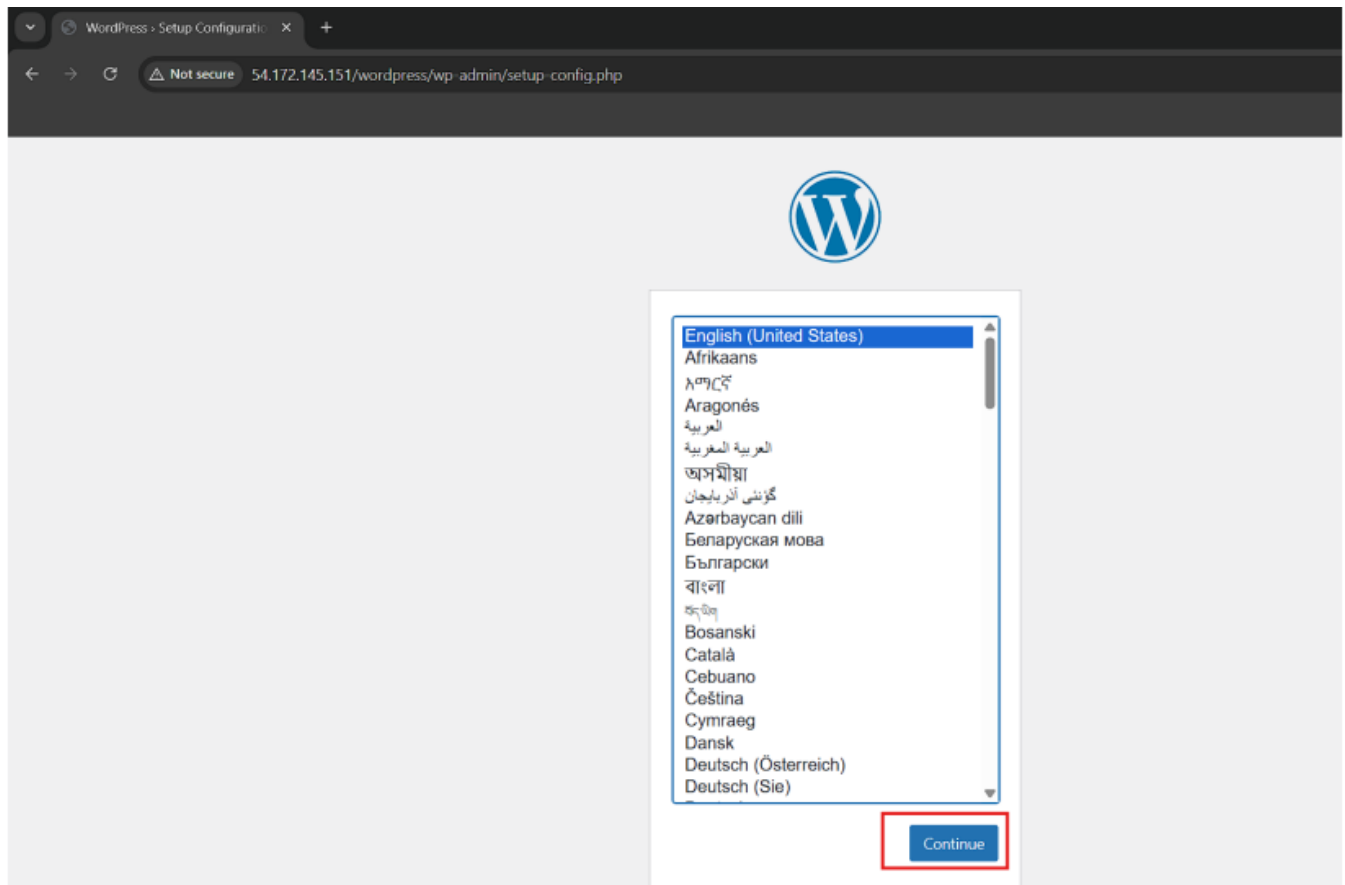
```
[ec2-user@ip-172-31-23-131 wordpress]$ sudo yum install php8.4-mysqlnd.x86_64
Last metadata expiration check: 0:17:42 ago on Sun Sep  7 08:28:43 2025.
Dependencies resolved.
=====
Package                                Architecture      Version
=====
Installing:
  php8.4-mysqlnd                        x86_64            8.4.10-1.amzn2023.0.1
Transaction Summary
=====
Install 1 Package
```

Step 8: Change ownership of the files

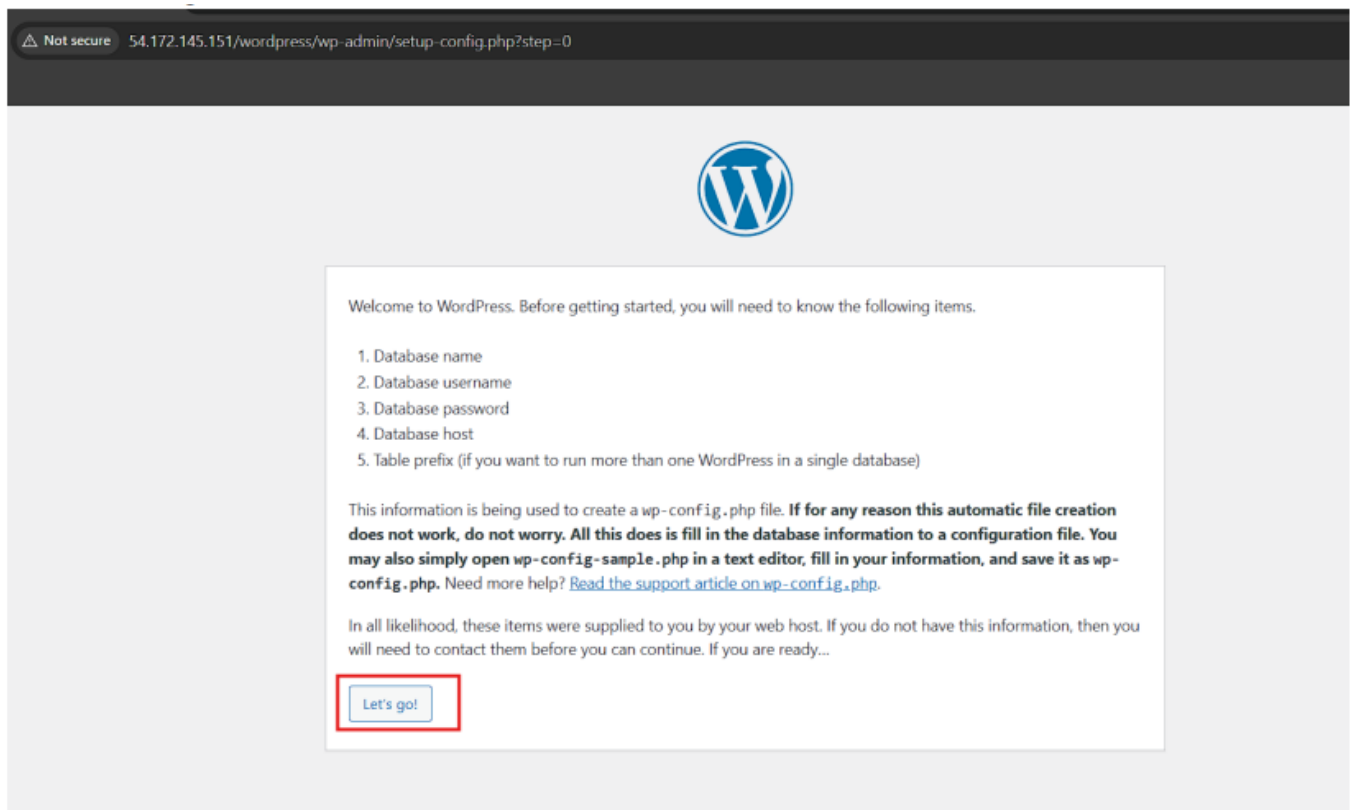
```
ec2-user@ip-172-31-23-131:/var/www/html
[ec2-user@ip-172-31-23-131 html]$ sudo chown -R apache:apache wordpress/
[ec2-user@ip-172-31-23-131 html]$ |
```

Step 9: Paste the Public IP and Paste it in any browser.

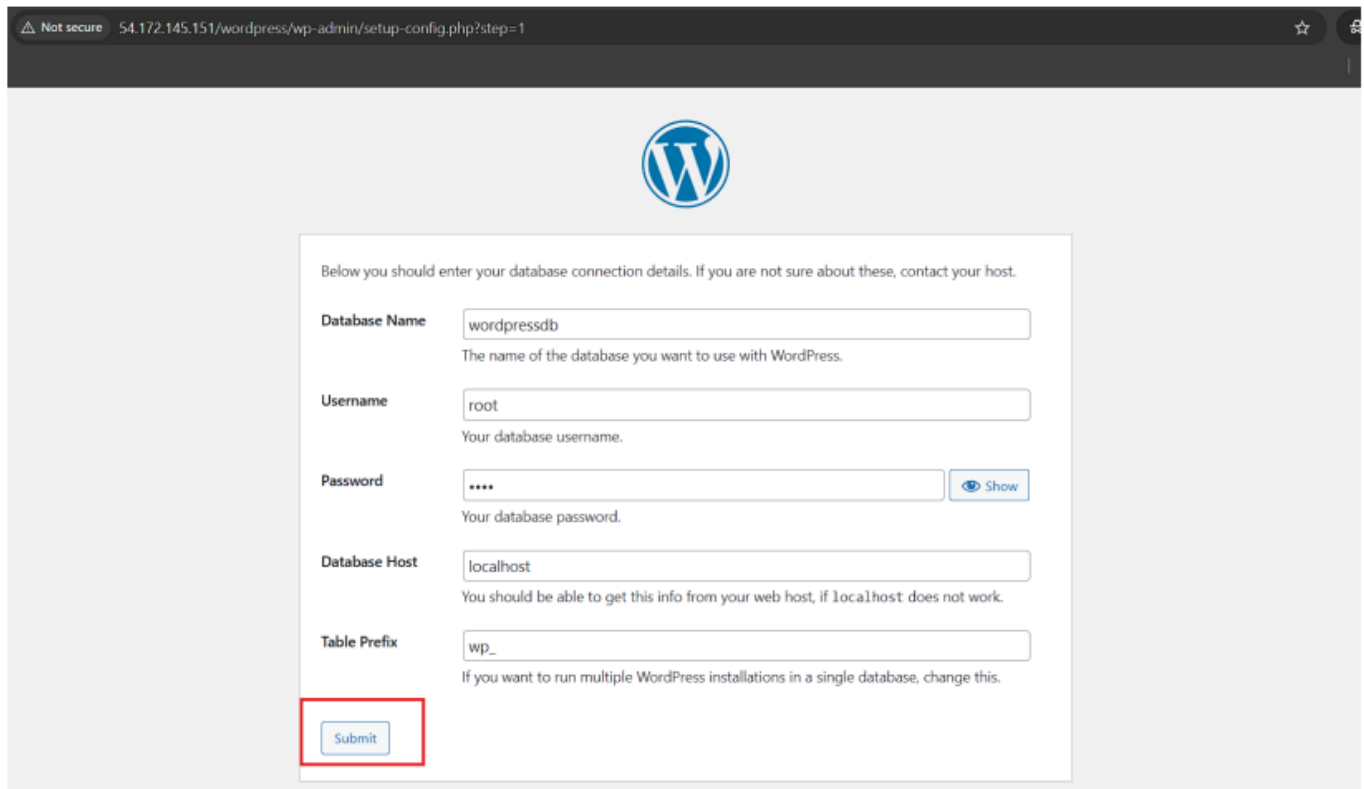
1. Click on Continue



2. Click on Let's go



3. Fill the information and click on Submit



The screenshot shows the WordPress installation configuration page. At the top, the browser address bar displays "54.172.145.151/wordpress/wp-admin/setup-config.php?step=1". The page features the WordPress logo and a form for database configuration. The form includes fields for Database Name (wordpressdb), Username (root), Password (masked with dots), Database Host (localhost), and Table Prefix (wp_). A "Submit" button is highlighted with a red rectangle.

Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name
The name of the database you want to use with WordPress.

Username
Your database username.

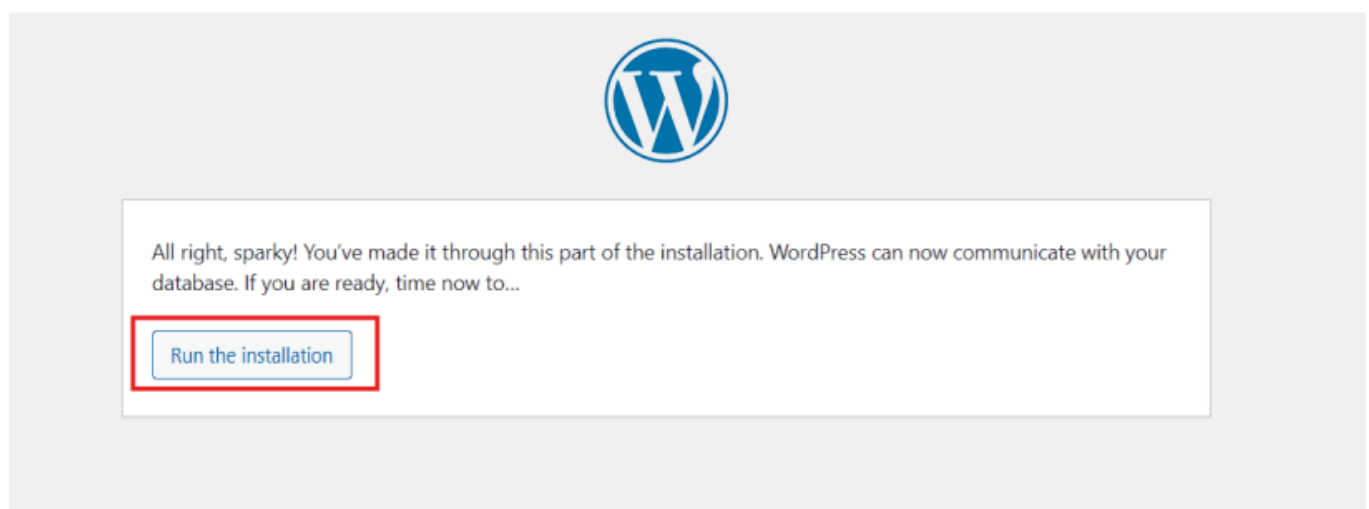
Password [Show](#)
Your database password.

Database Host
You should be able to get this info from your web host, if localhost does not work.

Table Prefix
If you want to run multiple WordPress installations in a single database, change this.

[Submit](#)

4. Run the Installation



The screenshot shows the WordPress confirmation screen after database configuration. It features the WordPress logo and a message: "All right, sparky! You've made it through this part of the installation. WordPress can now communicate with your database. If you are ready, time now to...". A "Run the installation" button is highlighted with a red rectangle.

All right, sparky! You've made it through this part of the installation. WordPress can now communicate with your database. If you are ready, time now to...

[Run the installation](#)

5. Fill the information and click on Install Wordpress

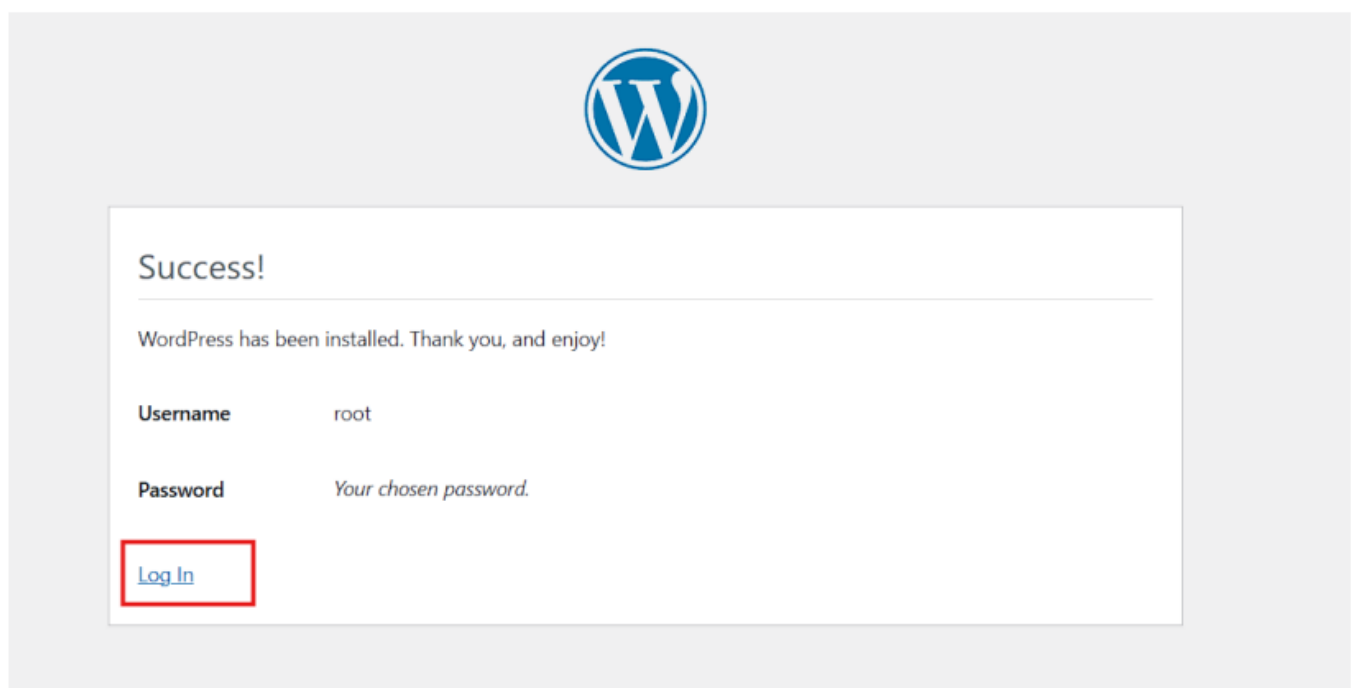
Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title	Tech Blog
Username	root Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.
Password	root Hide Very weak Important: You will need this password to log in. Please store it in a secure location.
Confirm Password	☒ Confirm use of weak password
Your Email	chaudharidhanashri2603@gmail.com Double-check your email address before continuing.
Search engine visibility	☒ Discourage search engines from indexing this site It is up to search engines to honor this request.

[Install WordPress](#)

6. Login to Wordpress



7. Deployed Wordpress successfully



Username or Email Address

Password

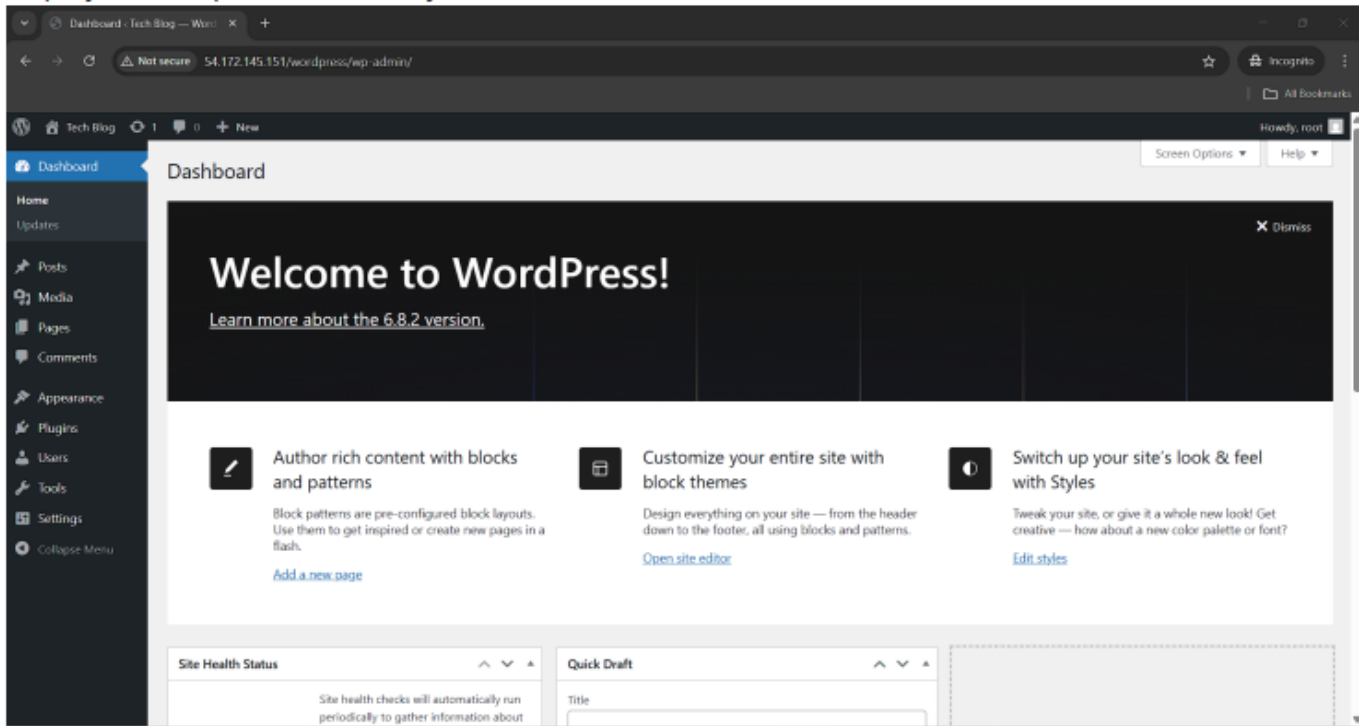
 

☐ Remember Me

[Lost your password?](#)

[← Go to Tech Blog](#)

8. Table automatically added to Database



Summary

This project is a complete guide to setting up a WordPress website on a Linux server. It uses Apache as the web server and MariaDB as the database. The guide explains how to download and install WordPress, set the right file permissions, create and connect the database, and configure the web server for live use. By following it, you'll get a secure, reliable, and fully working WordPress site that's ready for people to visit.